

“THIN FILMS CAN REDUCE ELECTRIC POWER AND UTILITY COSTS”



As the president of Applied Solar and chairman of the Applied Solar Council at Applied Materials, Dr Charles Gay is responsible for positioning Applied and the company's solar efforts with important stakeholders in the industry, technical community and governments, around the world. An industry veteran with over 30 years of experience in the solar industry, Dr Gay speaks to Uma Bansal, deputy editor, EFY, about the technological landscape for solar power and solar energy

■ DR CHARLES GAY, PRESIDENT, APPLIED SOLAR, APPLIED MATERIALS

How do you see solar market trends evolving over the next few years in India?

India has a favourable technological landscape for solar power and solar energy, and it is considered as one of the most lucrative sectors. Governments across the globe are also acknowledging the benefits of this technology. It is cost-effective for meeting peak power requirements like running air-conditioners or electrifying rural areas. Solar panels come with a 25-year warranty, which assures that they will generate at least 80 per cent or more of the power produced on their first day of manufacture. That is a significant accomplishment in the solar industry. Government policies are needed to assure stable, long-term predictability of the market.

There will be a growing demand for solar cells, modules and systems—both for export and domestic consumption. The semiconductor policy and the recent announcement of the National Solar Mission by the government of India reinforces country's commitment to the solar industry. Many state governments are also enabling ease of investments. Taken together, all this will further enable the growth and scaling of the associated solar ecosystem.

Crystalline silicon has the largest market share and thin film is a maturing technology, poised for high growth with its potential to reduce costs significantly.

Moser Baer Photovoltaic is now India's first manufacturer to implement Applied Materials' SunFab thin-film line for the mass production of solar modules. Tell us more about the benefits of this technology?

The thin-film line that Moser Baer has established with Applied Materials is a fully-integrated PV module production line that delivers leading-edge solar manufacturing capability using advanced engineering, process equipment, automation and other key supporting technologies. This

collaboration has set up the first advanced thin-film solar facility in India. The technology will provide innovative PV products that can significantly reduce electric power and utility costs and provide a cost-effective option for clean and renewable energy. Thin-film modules can facilitate reduction in the cost of energy generation for two reasons: One, thin-film amorphous silicon is more appropriate for warm climates. Two, large modules (5.7 m²) reduce the overall balance of system costs. The net effect is that the customer gets low-cost power.

Amorphous silicon has always been considered to give low efficiency. Please comment. Can you tell us which technologies are better?

Amorphous silicon is leading in the market due to its maturity. Cadmium Telluride (CdTe) and Copper Indium Diselenide (CIS) are also used for thin-film PV (TFPV). However, thin-film silicon can be easily processed as either amorphous or microcrystalline with widely differing optical properties, which allows a tandem or triple layer solar cell to be made with theoretical efficiencies far in excess of a single junction CdTe or CIS approach.

Applied Materials is committed to developing products for our customers based on their prevailing technology needs. We recognise the advantages of amorphous silicon as a low-cost material which is non-toxic and in abundant supply. In addition to the low cost, amorphous silicon in tandem has achieved 8 per cent efficiency in volume production with a clear path towards 10 per cent by 2010.

In view of the considerable fall in the silicon prices globally, don't you think TFPV is losing its price advantage too early? When do you think the impact of innovative approaches in the TFPV landscape be felt, possibly organic PV and dye sensitive cells?